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Wiley
International Transactions on Electrical Energy Systems
Volume 2025, Article ID 5946564, 12 pages
https://doi.org/10.1155/etep/5946564

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Research Article

A Critical Assessment of Cable Rating Methods Under Soil Drying Out Conditions

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Received 5 September 2024; Revised 2 June 2025; Accepted 29 August 2025

Academic Editor: Jesus E. Valdez-Resendiz

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The design of underground cable systems must account for the risk of soil drying out due to heat dissipation, which can degrade cable performance and lead to environmental concerns. This study investigates a cost-effective cable rating methodology tailored to South African conditions, where native soils are used instead of engineered backfill. Using the IEC 60287 standard, an Excel-based calculation tool is developed to assess the effects of key installation parameters, including soil thermal resistivity, ambient soil temperature and cable laying depth. Soil samples from Sandton, South Africa, revealed thermal resistivity ranging from 0.596 K·m/W, at 14.5% moisture, to 3.72 K·m/W, at 0% moisture, resulting in current ratings from 518.34 A to 224.21 A. Worst-case conditions—high resistivity, increased depth, 1150 mm and elevated soil temperature, 28°C—reduced ampacity by over 45%. The findings underscore the need to incorporate site-specific soil data and worst-case assumptions into cable rating designs to prevent thermal degradation. The developed method offers a practical, locally optimised alternative for utilities in semiarid regions.

Keywords: cable ampacity rating; electrical cable; power cable; soil drying out; soil thermal resistivity; South African conditions

1. Introduction

The novel generation power system requires diverse energy resources to support the reliability and security of the electric grid [1]. The electrical power cables, underground and overlines, suitably sustain the integration of diverse energy resources, which must handle the volatility of the current flow in the system. For the underground cable, the soil and environmental conditions play a major role on the efficient operation of the power cables. Therefore, there exists substantial research on the current rating of buried power cables and methods used to eliminate soil drying out due to the installed power cables [2]. The most common and effective method is to use engineered backfill material with low thermal resistivity to backfill the cable trenches in which the cables are buried [3, 4]. The engineered backfill material has been found to be expensive; hence, the use of South African native soil is being investigated, including all soil properties, such as thermal resistivity, composition, compaction and moisture content. Another reliable solution is the forced cooling method. This solution can be used provided there is sufficient cooling resource. The typical forced cooling systems include irrigation systems, separate pipe cooling, concrete racks cooling and forced air cooling [5–7]. The problem with the forced cooling method is the amount of water required to implement this solution. The scarcity of water in South Africa makes this solution not viable for South African use.

Some designs of cables use superconductors [8, 9]. These kinds of conductors use niobium or niobium-tin, which, at

temperatures within a few degrees of absolute zero, lose virtually all electrical resistivity [8]. Cooling with liquid helium is necessary to achieve these low temperatures [6]. These technologies have not matured enough and are very expensive. Such technologies are mostly considered for very long extra high voltage (EHV) cables, and thus, these technologies cannot be considered for medium voltage (MV) or low voltage (LV) applications. Furthermore, these technologies cannot be considered for South Africa due to the capital cost to procure these technologies.

The use of innovative technologies is the future of electrical networks [10]. Innovative technologies use fibre optic cable solutions to perform online monitoring and diagnostic analysis of the electrical infrastructure, thereby assisting utilities in planning their maintenance proactively [11]. The fibre optic cables can either be embedded in the different layers of the cable, during its manufacturing process, or they can be installed next to the power cable in the same trench [3]. The fibre optic cable will then be connected to an online monitoring system, e.g., distributed temperature sensing (DTS), conductor temperature monitoring technologies and dynamic current rating (DCR) technologies [12]. It is, however, not feasible to install these technologies in all MV and LV cable installations due to the cost of these technologies. The use of native soil for optimised cable systems design, which limits soil drying out, was investigated due to the cost of most of the technologies available. In [13], a modified algorithm strategy to deal with underground cables that can minimise material costs for high voltage (HV) networks is presented.

Besides, the majority of transmission and distribution networks in South Africa are overhead networks. The need to construct and install underground cable systems has increased due to the large-scale urban developments in South Africa over the past 50 years [14]. Thus, South African utilities must be equipped to design, install, operate and maintain these underground cable systems. Furthermore, the cable rating and design calculation methods [15] used for installing these cable systems need to be considered. Thus, this paper describes the procedure to be followed for optimised underground power cable systems designs for the consideration of soil drying out and dryer installation condition environments where the South African native soil will be used to backfill the cable tranches.

Soil drying out is a result of a moisture migration process which begins when a temperature gradient is imposed across the soil [16–18]. This temperature gradient will cause a water vapour pressure gradient to develop, resulting in moisture migrating away from the heat source. In this scenario, the heat source would be the buried cable. The moisture migration will eventually cause local drying of the soil near the cable [17, 19]. Thus, the original cable rating calculations [15], which would have been performed before the installation of the cable, would not be valid anymore since the resistivity of the soil would have changed.

Based on the assumptions made to perform the cable rating calculations, the calculated results for cable system designs may lead to the underutilisation of the installed expensive cable system infrastructures during operation by the power utilities. IEC 60287 specifies the constants and variables to be used for cable rating calculations [20–22]. This is mostly based on different temperature levels. The cable-specific constants and variables are dependent on the construction, materials and method used for the manufacturing of the cable [23, 24]. However, the onsite dependent constants and variables (for the cable installation conditions) may be unknown at the initial design phase for final cable rating calculations.

This paper describes the procedure to be followed for a suitable current rating method that prevents soil drying out by using the IEC 60287 calculation method on South African native soil [25]. A contrast has been observed with the existing techniques regarding the native soil and local climate, which have been underexplored in several prior works. Therefore, a procedural guideline for the utilities is presented. This includes choosing or sizing the required cable based on load requirements and available manufacturer's data sheets, performing soil thermal resistivity tests on the selected cable route, determining cable laying depth depending on the chosen cable route, and performing soil ambient temperature measurement and cable laying depth. Then, the cable current rating calculations will be performed to verify if the selected cable size will be sufficient for the load; otherwise, a bigger cable size might need to be chosen.

The contribution of the research study can be listed as follows:

- A developed cable rating calculation tool that considers three worst-case scenarios. This study assesses by considering three important aspects of underground cable deployment, including soil resistivity, ambient temperature and laying depth. The assessment is made under real-world environmental conditions in Sandton, South Africa.
- A cable current rating procedure has been developed. This procedure provides utilities with a simplified guideline for performing cable rating calculations on underground cables in order to prevent soil drying.
- Implementing the developed procedure and cable rating calculation tool will prevent soil from drying out during cable installation, avoiding costly engineered backfill.

The remaining part of this research study can be summarised as follows: Section 2 details the calculation method based on IEC 60287. Section 3 presents different design parameters of cables based on the South African climate. Section 4 assesses the factors affecting T_4. Sections 5 presents different results and discussion. Finally, Section 6 concludes the research work and proposes future research.

2. System Methodology

IEC 60287 was used to perform cable rating calculations on three different cable manufacturers' data sheets. The design parameters of the cable used were as follows: 185 mm^2, 3 cores, copper conductors, cross-linked polyethylene (XLPE) insulated and steel wire armoured; the cable has water

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blocking tape and has polyethene outer sheath. Figure 1 shows a typical cross-section of a 3-core cable. The functions of the different layers of the cable are as follows:

- Annealed stranded copper conductor to carry current,
- Semiconductive conductor screen to provide a smooth and radial electric field within the insulation,
- XLPE insulation layer for insulation purposes,
- Semiconductive core screen smoothens and contains the electric field in the insulation of the cable,
- Water blocking semiconductive tape to contain electric field and to block water from travelling longitudinally in the cable and water ingress should be there along the cable,
- Copper-screened tape to perform secondary earthing of the cable once installed,
- Filler: to fill up the gaps between the three cores of the cable and assist in creating a round cable,
- Binding tape to bind the three cores of the cable and filler together,
- PVC bedding is a cushioning layer for the armouring, thereby preventing the sharp edges of the armouring from puncturing the XLPE,
- Galvanised steel wire armouring provides a path for the flow of charging current and fault current. It also provides mechanical protection of the cable,
- Binding tape binds all the layers of the cable, including the armouring,
- UV-stabilised PVC sheath jacket for covering the cable, and it provides additional mechanical protection to the cable.

The cable current rating calculations were performed based on the different layers, as stated above. An Excel calculating tool based on IEC 60287 is developed and used to assess the cable rating.

The South African native soil was to be used to backfill the cable trenches. Focus was directed onto the assumptions made for the calculations of T_4. The maximum allowed cable conductor temperature was 70°C. The ideal installation conditions considered were as follows:

- Directly buried 850 mm underground,
- Soil thermal resistivity was 1.2 K·m/W, and
- Soil ambient temperature was 25°C.

The calculated cable rating in accordance with IEC 60287 for the ideal cable installation conditions, as stated above, is shown in Table 1.

Equation (1) was used to calculate the current rating values in Table 1. The parameters and values for constants for Equation (1) were obtained in IEC 60287.

$$I = \left[\frac{\Delta\theta - W_d\left[0.5T_1 + n\left(T_2 + T_3 + T_4\right)\right]}{RT_1 + nR\left(1 + \lambda_1\right)T_2 + nR\left(1 + \lambda_1 + \lambda_2\right)\left(T_3 + T_4\right)}\right]^{0.5}, \tag{1}$$

where I, R, $\Delta\theta$, W_d and n are the current flowing in one conductor (A), alternating current resistance per unit length of the conductor at maximum operating temperature (Ω/m), conductor temperature rise above the ambient temperature (°K), dielectric loss per unit length for the insulation surrounding the conductor (W/m) and number of load-carrying conductors in the cable (conductors of equal size and carrying the same load); λ_1 and λ_2 are, respectively, the ratio of losses in the metal sheath to total losses in all conductors in that cable and in armouring total losses in all conductors in that cable; T_1, T_2, T_3 and T_4 are, respectively, thermal resistance per unit length (°K·m/W) between one conductor and the sheath, of the bedding between sheath and armour, of the external serving of the cable and between the cable surface and the surrounding medium.

Figure 2 shows all the different thermal resistances on the various layers of the cable. T_1, T_2 and T_3 depend on the cable design and raw material used to manufacture the cable, and they can be calculated using the applicable equations given in IEC 60287. However, T_4 is dependent on the onsite installation conditions, which may be unknown at the initial design phase for final cable rating calculations.

The onsite installation conditions include the cable burial depth, the type of soil used to backfill the cable trench, the moisture content of the soil, the number of cable circuits in the trench and the space between the different cable circuits. For this research, the cable will be directly buried 850 mm underground, the cable trench will be backfilled with native soil and there will only be one cable circuit in the trench.

Different cable designs and installations have been considered in IEC 60287; the various designs, as stated in IEC 6287, include the following:

1. The different insulation materials: This could be paper-insulated lead-covered cable (PILC), XLPE and oil-filled cable.
2. The design of cable: whether single core or three cores.

For this research, only the equations for XLPE and three core cables will be considered. This is due to the fact that Eskom has standardised XLPE cables for MV applications. The equations specified below are based on 60,287 [15].

Equation (2) is used to determine the conductor temperature rise above the ambient temperature of any cable and any design.

$$\begin{aligned}\Delta\theta = &\left(I^2R + \left(\frac{1}{2}\right)W_d\right)T_1 + \left[I^2R\left(1 + T_1\right) + W_d\right]nT_2 \\ &+ \left[I^2R\left(1 + T_1 + T_2\right) + W_d\right]n\left(T_3 + T_4\right).\end{aligned} \tag{2}$$

3. South African Climate

South Africa has been going through a desertification process over the past few years. Thus, South Africa can be considered as a semidessert country, where the majority of the country gets rainfall between October and March, with an average annual rainfall of 650 mm. The cable current

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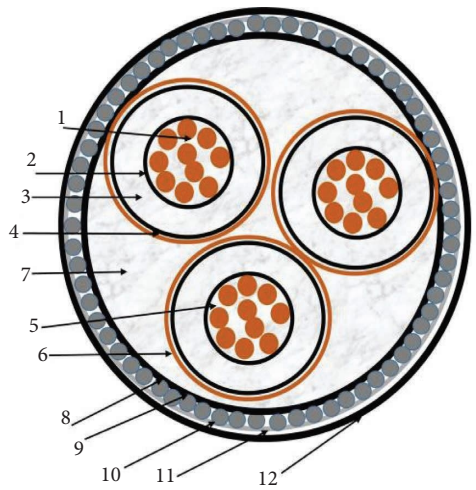

1 Annealed stranded copper conductor
2 Semiconductive conductor screen
3 XLEP Insulation
4 Semiconductive core screen
5 Semiconductive water blocking tape
6 Copper screened tape
7 Filler
8 Binding tape
9 FR PVC bedding
10 Galvanized steel wire armouring
11 Binding tape
12 UV stabilized PVC sheath

Figure 1: Typical cross-sectional drawing for MV XLPE.

Table 1: Cable current rating in ideal cable installation condition.

Cable manufacturer's data sheet used	Calculated current rating
Cable A	382.371 A
Cable B	382.869 A
Cable C	381.143 A

ratings were performed for different installation conditions, taking into consideration soil thermal resistivity, soil ambient temperature and the cable laying depth.

3.1. Cable Rating in a Nonideal Cable Installation Environment

3.1.1. Current Rating Based on the Change in Soil Thermal Resistivity. The use of native soil to backfill cable trenches will require that thermal resistivity tests be performed. For this research, soil samples were taken from a known location in South Africa (Sandton), where an electrical cable was going to be installed. The soil thermal resistivity measurements were performed on the South African native soil at different moisture content, which in turn resulted in different soil thermal resistivity. Table 2 presents the soil thermal resistivity in the function of soil moisture. The cable current rating calculations, as shown in Table 3, were then performed for the measured soil thermal resistivity described in Table 2.

The developed Excel calculating tool was used to calculate the cable rating; only the soil thermal resistivity was changed, and the calculated current rating was recorded in Table 3. Table 3 shows the cable rating calculation results based on different soil thermal resistivity. The soil thermal resistivity test results, as stated in Table 2, were used to recalculate the current rating of the three different cables. It is shown in Table 2 that the higher the soil moisture content, the lower the thermal resistivity of the soil. The best current rating results were obtained when the soil thermal resistivity was 0.596 K·m/W, and moisture content was 14.5%.

The worst soil thermal resistivity at 0% moisture content should be used for cable current rating to eliminate soil drying out due to the installed power cables.

3.1.2. Cable Current Rating Based on Different Soil Ambient Temperature. The South African average soil ambient temperature at about 850 mm below ground is about 25°C. The average soil ambient temperature is expected to increase due to global warming and other factors, such as installed services underground. The more services installed underground, the greater will be the soil ambient temperature, and the more difficult it will be to dissipate heat. The ideal soil ambient temperature at about 850 mm below ground is about 25°C. Thus, cable current rating was performed for different soil ambient temperatures (Table 4), assuming that the soil ambient temperature will be increasing over the next few years. The developed Excel calculating tool was used to calculate the cable rating; only the soil ambient temperature was changed, and the calculated current rating is recorded in Table 4. The best current rating results were obtained when the soil ambient temperature was 24°C.

The worst soil ambient temperature in Table 4 (28°C) illustrates that the higher the soil ambient temperature, the worse the cable current rating is. Therefore, it is recommended that the soil ambient temperature at cable laying depth should be measured and used in the cable current rating calculations.

3.1.3. Cable Current Rating Based on Different Cable Laying Depths. The onsite installation conditions differ per project site. There could be site conditions that would require that the cable trench be dug deeper or shallower than the ideal installation depth. The shallow cable trench could pose an environmental hazard or a safety risk. The environmental hazard could be due to soil erosion, which could, in turn, expose the cable to the public. The safety of human beings and animals could be compromised once the cable is exposed to the public. Thus, power cables are predominately buried deeper than the ideal conditions to ensure the safety of the public and animals. Another installation scenario that

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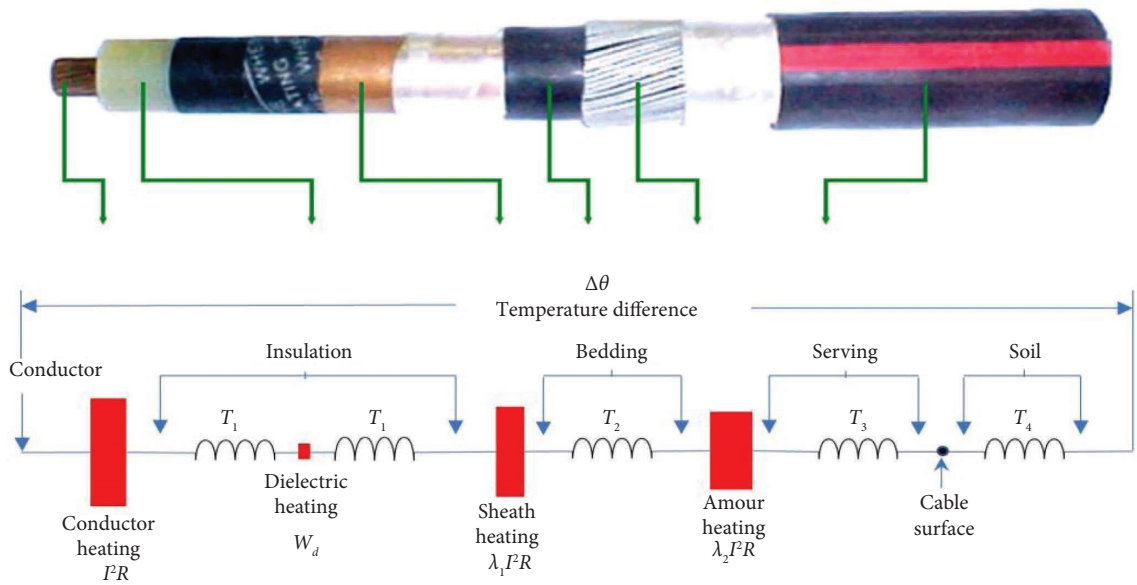

Figure 2: Single core cable showing all the different thermal resistivity on the various layers of the cable.

Table 2: Measured soil thermal resistivity at different moisture content.

Soil moisture content (%)	Soil thermal resistivity (K·m/W)
14.5	0.596
0	3.720
2	2.341
5	1.646

could result in deeper cable trenches could be other services being installed in the same servitudes.

With all these scenarios, the standard depth for the cable trench will not be met; this, in turn, will affect the cable rating of cables which will be installed in those trenches; hence, cable rating due to different cable trench/laying depths was investigated (Table 5). The following cable trench depths were considered for the purpose of this research: 750, 850, 950, 1000 and 1150 mm.

The developed Excel calculating tool was used to calculate the cable rating; only the cable laying depth was changed, and the calculated current rating is recorded in Table 5. The best current rating results were obtained when the cable laying depth was 750 mm.

Due to the uncertainty of the cable laying depth, it is recommended that the utilities perform the cable current rating calculations again to verify the initial calculation once the cable laying depth has been confirmed and verified.

3.1.4. Current Rating Based on Worst-Case Installation. The probability of the three worst-case scenarios happening at the same time and location was taken into consideration, where the soil moisture content was 0% with soil thermal resistivity of 3.72 °K·m/W, the soil ambient temperature was 28°C and the cable laying depth was 1150 mm. Table 6 shows the cable current rating where the three nonideal conditions were taken into consideration.

The developed Excel calculating tool was used to calculate the cable rating. The soil thermal resistivity, the cable laying depth and the soil ambient temperature listed in Table 6 were used to perform the cable current rating calculations. The current rating reduced from 382.37 A to 208.70 A (for Cable A); this resulted in a reduction of about 45.4% of the calculated current rating with ideal conditions.

4. Factors Affecting T_4

The relationship between T_4 and the nonideal installation conditions was investigated. The behaviour of T_4 was analysed for different installation conditions; this includes the change in soil thermal resistivity, the change in soil ambient temperature and the change in cable burial depth.

4.1. T_4 Based on the Change in Soil Thermal Resistivity. The following results were observed when the soil thermal resistivity was changed: the permissible temperature rise of the conductor above soil ambient temperature ($\Delta\theta$) remained the same, and the current rating of the cable and T_4 reduced when the soil thermal resistivity was increased, as shown in Tables 7, 8, and 9.

Figure 3 is a graphical representation of the results stated in Table 7 above. It is shown in Figure 3 that the current rating of the cable is inversely proportional to T_4, and inversely proportional to the increase in soil ambient temperature; however, there is no mathematical effect between the changes in soil thermal resistivity and the temperature rise of the cable [1].

4.2. T_4 Based on a Change in Soil Ambient Temperature. The permissible temperature rise of the conductor above soil ambient temperature ($\Delta\theta$) and the current rating reduced for an increase in soil ambient temperature while T_4 remained the same.

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Table 3: Calculated cable current rating at different soil thermal resistivity.

Cable manufacturer	Cable A	Cable B	Cable C	Average
I (at 14.5% & 0.596 K·m/W)	518.746 A	519.540 A	516.723 A	518.34 A
I (imported soil & 1.2 K·m/W)	382.371 A	382.869 A	381.143 A	382.13 A
I (at 5% & 1.646 K·m/W)	330.663 A	331.070 A	329.667 A	330.47 A
I (at 2% & 2.341 K·m/W)	280.169 A	280.495 A	279.372 A	280.01 A
I (at 0% & 3.72 K·m/W)	224.330 A	224.582 A	223.730 A	224.21 A

Table 4: Current rating based on the change in soil ambient temperature.

Cable manufacturer	24°C	25°C	26°C	27°C	28°C
Cable A	386.597 A	382.371 A	378.099 A	373.777 A	369.405 A
Cable B	387.100 A	382.869 A	378.590 A	374.263 A	369.886 A
Cable C	385.355 A	381.143 A	376.884 A	372.577 A	368.219 A

Table 5: Cable current rating based on different cable laying depth.

Cable manufacturer (mm)	Cable A	Cable B	Cable C
750	382.321 A	388.833 A	387.031 A
850	382.371 A	382.869 A	381.143 A
950	377.308 A	377.793 A	376.132 A
1000	375.040 A	375.520 A	373.888 A
1150	369.062 A	369.527 A	367.967 A

Table 6: Current rating based on worst installation condition.

Installation condition	Value	Cable A	Cable B	Cable C
Worst installation condition (I_{worst})				
Soil thermal resistivity	3.72 K·m/W	208.70 A	208.93 A	208.19 A
Cable laying depth	1150 mm			
Soil temperature	28°C			
Ideal installation condition (I_{ideal})				
Soil thermal resistivity	1.2 K·m/W	382.37 A	382.87 A	381.14 A
Cable laying depth	850 mm			
Soil temperature	25°C			
$I_{rating} = I_{worst}/I_{ideal}$	Average: 45.40%	45.42%	45.43%	45.38%

Table 7: T_4 when the soil thermal resistivity is changed.

	Cable A		
Soil thermal resistivity (K·m/W)	ΔΩ (°C)	I (A)	T_4 (K·m/W)
0.6	45	517.3085	0.356034930
0.8	45	457.8175	0.474713241
1.0	45	415.0372	0.593391550
1.2	45	382.3714	0.712069870
1.4	45	356.3748	0.830748172
1.6	45	335.0487	0.949426482
1.8	45	317.1435	1.068104790
2.0	45	301.8327	1.186783102
2.2	45	288.5448	1.305461413
2.4	45	276.8702	1.068104790
2.6	45	266.5070	1.186783102

Table 8: T_4 when the soil ambient temperature is increased.

	Cable A		
Ambient temperature (°C)	$\Delta\theta$ (°C)	I (A)	T_4 (K·m/W)
23.0	47.0	390.777	0.71207
23.5	46.5	388.692	0.71207
24.0	46.0	386.597	0.71207
24.5	45.5	384.490	0.71207
25.0	45.0	382.371	0.71207
25.5	44.5	380.241	0.71207
26.0	44.0	378.099	0.71207
26.5	43.5	375.944	0.71207
27.0	43.0	373.777	0.71207
27.5	42.5	371.597	0.71207
28.0	42.0	369.405	0.71207

Table 9: Relationship between the change in cable laying depth, T_4, temperature rise and current rating.

	Cable A		
Cable laying depth (m)	$\Delta\theta$	I (A)	T_4 (K·m/W)
0.650	45	395.47736	0.660756803
0.70	45	391.72211	0.674936407
0.75	45	388.32101	0.688134063
0.80	45	385.21871	0.700477184
0.85	45	382.37148	0.712069861
0.90	45	379.74422	0.722998235
0.95	45	377.308343	0.733334400
1.00	45	375.04030	0.74313930
1.05	45	372.92052	0.75246495
1.10	45	370.93251	0.76135604
1.15	45	369.06230	0.76985130

Figure 3 is a graphical representation of the results stated in Table 8. It is shown in Figure 4 that the current rating of the cable and temperature rise is inversely proportional to the increase in soil ambient temperature; however, there is no mathematical effect between the value of T_4 and the increase in soil ambient temperature [15].

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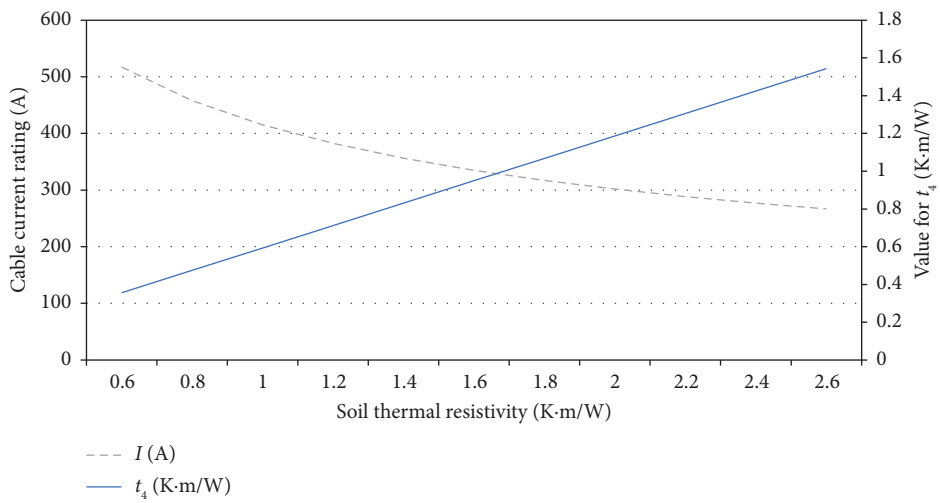

Figure 3: T_4 and the cable current rating when the soil thermal resistivity is changed.

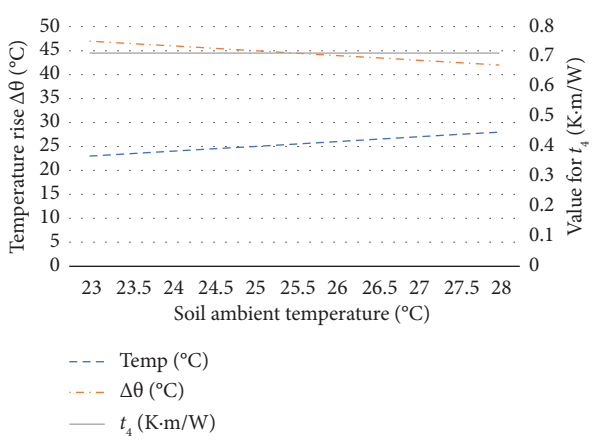

Figure 4: T_4 when the soil ambient temperature is increased.

4.2.1. T_4 Based on a Change in Cable Laying Depth. The current rating of the cable was reduced for an increase in cable laying depth, as shown in Table 5; the value of T_4 increased while the soil ambient temperature remained unchanged when the cable laying depth was increased, as detailed in Table 9.

Figure 5 is a graphical representation of the results stated in Table 9. It is shown in Figure 5 that the current rating of the cable and T_4 are inversely proportional to the increase in cable laying depth. At the same time, T_4 is directly proportional to the increase in cable laying depth.

5. Representation of Results

5.1. Results Based on Increased Soil Thermal Resistivity. The thermal resistivity of the South African native soil is expected to increase due to global warming and desertification. The desertification phase will result in the migration of the moisture content in the soil; this, in turn, will result in higher soil thermal resistivity. The higher the soil's thermal resistivity, the more difficult it will be for the soil to dissipate heat around the cable. An increase in the conductor temperature will increase the surface temperature of the cable; consequently, the soil closest to the cable will solidify depending on the surface temperature of the cable. Thus, the need to perform soil thermal resistivity measurements before installing power cables is justified. The soil thermal resistivity measurement performed on the soil sample taken from the Sandton area (in South Africa) proved the impact of change in moisture content in the soil. The measured soil thermal resistivity increased from 0.596 K·m/W when the soil moisture content was 14.5% to 3.720 K·m/W when the soil moisture content was 0%. The calculated average current rating due to the two different soil thermal resistivity was 518.34 A and 224.21 A, respectively.

5.2. Results Based on Soil Ambient Temperature. Global warming will further influence the increase of soil ambient temperature. The average ambient temperature in South Africa has been increasing by an average of 0.9°C [26] per year for the past few years due to global warming; this equates to an average increase of approximately 9°C every 10 years. This could result in a fraction of an increase in soil ambient temperature. This, in turn, will have a negative impact on the ampacity of buried cables.

Tables 4 and 8 show that the current rating calculations reduce by an average of 4.27 A (1.12%) for a 1°C increase in soil ambient temperature.

5.3. Results Based on the Change in Cable Laying Depth. The onsite installation conditions differ per project site. There could be site conditions that would require that the cable trench be dug deeper or shallower than the ideal cable laying depth of 850 mm. These changes in cable laying depth have a great impact on the cable ampacity rating. The deeper the cable is laid, the worse the cable ampacity rating.

5.4. Factors Affecting T_4. The behaviour of T_4 under different installation conditions was analysed: this included the change in soil thermal resistivity, the change in soil ambient

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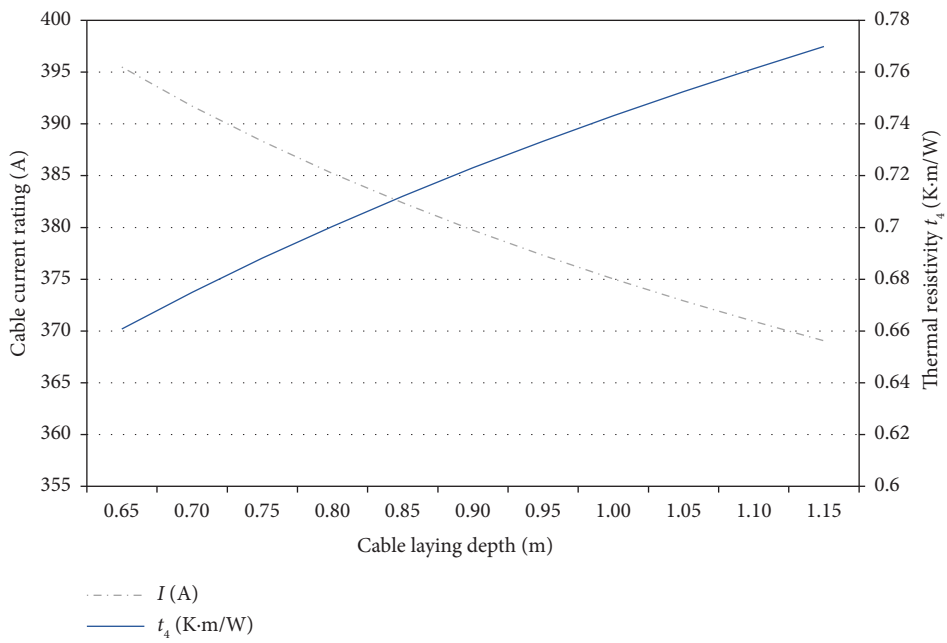

Figure 5: T_4 when cable laying depth is changed.

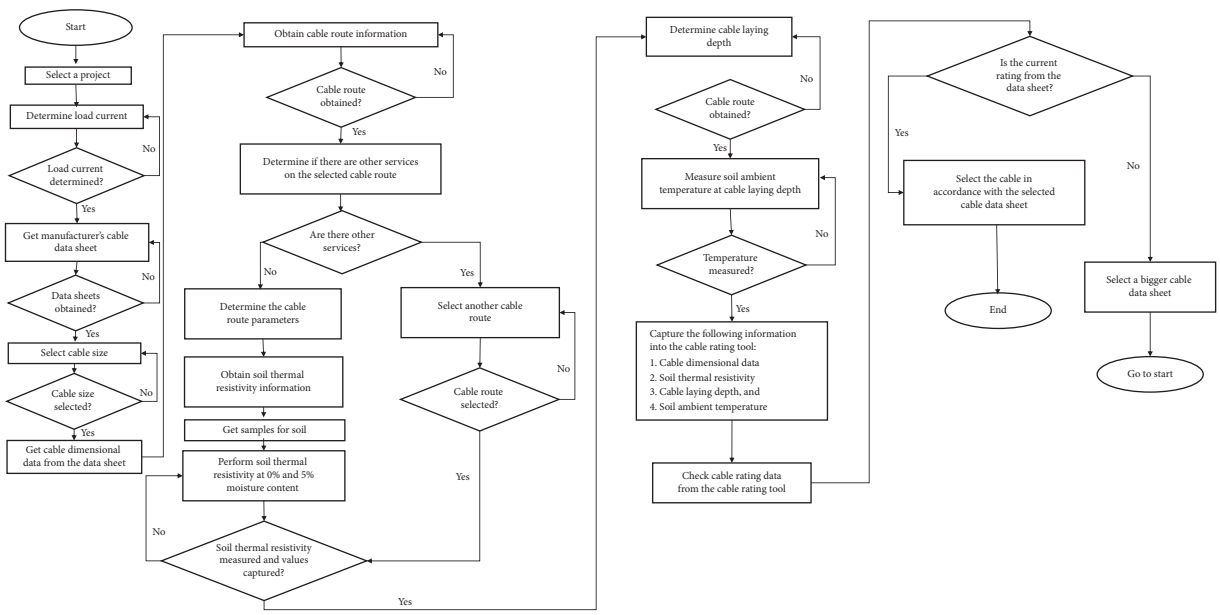

Figure 6: Procedure to be followed for cable rating calculations when using the cable rating tool.

temperature and the change in cable laying depth. It was found that there is no mathematical relationship between the change in the soil ambient temperature and T_4, nor does a mathematical relationship exist between the cable laying depth and the temperature rise of the cable. This proves that the thermal resistivity of the soil increases the deeper trench. Therefore, the thermal resistivity of soil tends to increase with burial depth due to reduced moisture content and increased compaction in deeper soil layers, which impedes heat dissipation from buried cables [17, 19]. This relationship was validated in this study, where cable ampacity decreased, and thermal resistance, T_4, increased with greater laying depth. This may be attributed to increased soil compaction and reduced moisture retention at depths around 1000 mm, which can elevate thermal resistivity and hinder effective heat dissipation from buried cable.

5.5. Procedure to Be Followed for Cable Current Rating. Figure 6 shows a procedure that utilities can use to perform current cable rating calculations. The procedure aims to assist the utilities and Eskom in performing cable current rating calculations that will eliminate soil drying out due to installed cables.

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5.6. Discussion. In this paper, it was shown that three major installation conditions/parameters are to be considered when performing current rating calculations for a single circuit of buried cables; these parameters are the soil thermal resistivity, including the soil moisture content, the cable laying depth and the soil ambient temperature. These three parameters are unfortunately not fixed as they can change depending on the environment and installation condition.

It was shown in this paper that the best current rating for buried cables is achieved when the soil is moist. The more humid the soil, the less is the soil thermal resistivity, and the more current can be transmitted through a cable; thus, the current rating is directly proportional to the percentage of moisture content of the soil where the cable will be installed. South Africa has been going through a desertification phase due to global warming. This means that the design engineers will have to be more conservative when designing and selecting cable sizes for projects.

The average ambient temperature has been increasing over the years; this is directly proportional to the soil ambient temperature to which power cables are installed.

Utilities have standard designs for cable trenches for different types of cable installations. However, it is not always possible to comply with the required installation depths due to onsite conditions. Some installations would require deeper cable trenches compared to the standard design of cable trenches. Derating of the cable will then have to be considered for these kinds of installations. This paper has shown that the deeper the cable laying depth, the less the current rating. The best results were achieved when the cable was laid in a shallow cable trench. This, however, cannot be permitted as these cables would be alive, carrying current in the range of hundreds of amps and the range of thousands of volts; this is a safety hazard to the public and the environment.

The cable rating results deteriorated significantly when all three nonideal parameters—soil thermal resistivity of 3.72 K·m/W, cable laying depth of 1150 mm and soil ambient temperature of 28°C—were simultaneously considered. Under these worst-case conditions, the calculated current rating for Cable A dropped to 208.70 A, compared to 382.37 A under ideal conditions (1.2 K·m/W thermal resistivity, 850 mm depth and 25°C soil temperature), resulting in an average ampacity reduction of approximately 45.4%. This reduction is primarily due to the influence these parameters have on the thermal resistance to the surrounding environment (denoted as T_4 in the IEC 60287 standard). According to the ampacity formulation, in Equation (1), the allowable current I is inversely related to the square root of the total thermal resistance, which includes T_4. In this circumstance, T_4, the thermal resistance between the cable surface and surrounding soil increases significantly with higher soil thermal resistivity, greater burial depth and increased ambient temperature.

For example, T_4 increases from 0.712 K·m/W, at ideal 1.2 K·m/W soil resistivity, to over 1.186 K·m/W at 3.72 K·m/W soil resistivity, as presented in Table 7. This increase in T_4 directly raises the denominator of the ampacity formula, reducing the current rating. Moreover, while $\Delta\theta$ was held constant at 45°C in all simulations to isolate the impact of T_4, the compounded effect of reduced heat dissipation (from poor thermal conductivity), increased soil temperature and deeper burial, which typically corresponds to denser, less porous soil layers. This amplifies the thermal bottleneck around the cable. Therefore, the study confirms that worst-case environmental conditions must be assumed during the design phase to avoid dangerously overestimating cable ampacity. This also justifies the recommendation to use the lowest practical soil moisture condition, considering 0% moisture content, equating to the highest resistivity, for sizing underground cables in semiarid regions such as South Africa, as well as the incorporation of real-time monitoring for instant DTS systems in critical installations. Besides, it should be noted that the manufacturers' data sheets shown in Figure 1 were used to perform the calculated results, as shown in Table 10.

It was found that the current rating of the cable is inversely proportional to T_4, inversely proportional to the increase in soil ambient temperature; however, it was found that there is no mathematical relationship between the changes in cable laying depth and the temperature rise of the cable. The temperature rise of the cable due to increased cable depth can only be verified by measuring the ambient temperature at the cable laying depth. However, the probability of an increase in temperature rise is high when the cable laying depth is increased. This, however, proves that the thermal resistivity of the soil increases the deeper trench, depending on moisture content, moisture content, soil composition and layering and soil composition and layering. This may be attributed to the compacted and denser nature of deeper soil layers, which are often less porous and retain less moisture—factors that collectively reduce the soil's ability to dissipate heat away from the buried cable.

The flow diagram of the procedure, as shown in Figure 6, is aimed at assisting the utilities in performing the cable rating calculations. The tool can further be used to verify the work done by consultants (where applicable).

The use of DTS to monitor the surface temperature of the cable outer sheath against the allowed maximum surface temperature of the cable outer sheath would reduce the probability of soil drying out due to the high surface temperature of the cable outer sheath. This is because the DTS will allow the utilities to proactively perform the necessary switching to reduce the load on the affected cable networks.

Although there are available technologies [3, 12, 27, 28] that can improve cable ampacity rating, procuring the relevant technologies is difficult in poor countries due to a lack of funds. In such countries, it is prudent to use the native soil to backfill the cable tranches to minimise the capital cost for the installation of underground cable systems.

The current rating calculations proved that the drier the soil, the worse the current rating of the installed cable [6, 29–32]. Therefore, designers must consider the worst-case scenario when performing current ratings for buried

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Table 10: Lists of material constants used to perform the calculations.

Symbol	Description	Units	Constant value
$\alpha_{20\,\mathrm{Arm}}$	Temperature coefficient of electrical resistivity for steel armour at 20°C per Kelvin	/K	0.0045
$\alpha_{20\,\mathrm{Cu}}$	Temperature coefficient of electrical resistivity for copper conductor at 20°C per Kelvin	/K	0.00393
$\alpha_{20\,\mathrm{Cu\,tap}}$	Temperature coefficient of electrical resistivity for metallic tape (copper tape) at 20°C per Kelvin	/K	0.004
ε	Relative permittivity of insulation	—	2.5
f	System frequency	Hz	50
k_p	Proximity effect factor	—	1
k_s	Skin effect factor	—	1
ρ_{armou}	Thermal resistivity of the armouring (Steel)	K·m/W	0.000000138
$\rho_{\mathrm{Cu\,tape}}$	Thermal resistivity of copper tape at 20°C	K·m/W	0.000000214
ρ_i	Thermal resistivity of insulation material (XLPE)	K·m/W	3.5
ρ_{fillers}	Thermal resistivity of fillers	K·m/W	6
ρ_{sheath}	Thermal resistivity of cable outer sheath material (PE)	K·m/W	3.5
ρ_{sc}	Thermal resistivity of semiconductive layer	K·m/W	2.5
$\rho_{\mathrm{water\,b/t}}t$	Thermal resistivity of water blocking tape	K·m/W	2.5
π	Pi	—	3.1415926536
$\tan\,\delta$	Insulation loss factor	—	0.004
$\lambda 1''$	Eddy current losses (this value is zero for 3 core cables)	—	0.00

cables to prevent soil from drying out in the near future. These scenarios include soil thermal resistivity, soil moisture content, cable laying depth and soil ambient temperature.

6. Conclusion

This study has demonstrated that soil conditions, particularly thermal resistivity, ambient temperature and cable laying depth, significantly influence underground cable ampacity. The experimental data collected from native South African soil, specifically from Sandton, revealed a wide range of soil thermal resistivity values, from 0.596 K·m/W at 14.5% moisture to 3.72 K·m/W at 0% moisture, with corresponding current ratings decreasing from 518.34 A to 224.21 A. It was shown that soil drying out drastically reduces cable performance, and therefore, worst-case soil thermal resistivity should be considered in the design to avoid long-term degradation. In this study, the limitations of findings are based on measurements taken from a single soil type within a specific South African locality. Soil composition, moisture retention, compaction and ambient temperature vary significantly across regions and climates. Therefore, the applicability of the proposed cable rating methodology and Excel-based calculation tool may be limited in areas with differing soil textures, e.g., sandy, clay, or loamy soils, different seasonal rainfall patterns, or cooler climates.

Furthermore, this work assumes constant environmental parameters during operation, which are practically variable. Besides, transient conditions, such as heavy rainfall, drought or anthropogenic heat sources, e.g., nearby buried infrastructure, could alter soil properties over time. Therefore, future research should explore soil thermal dynamics over longer periods, assess various native soils from different geographical zones and consider the effects of climate change. It would also be valuable to integrate economic assessments and real-time monitoring systems for adaptive cable management strategies under uncertain and evolving environmental conditions.

Nomenclature

Parameters and Constants

DCR Dynamic current rating
DTS Distributed temperature sensing
EHV Extra high voltage
FR Flame retardant
HV High voltage
IEC International Electrotechnical Commission
LV Low voltage
MV Medium voltage
PE Polyethylene
PILC Paper-insulated lead-covered cable
PVC Polyvinyl chloride
UV Ultraviolet
XLPE Cross-linked polyethylene

Data Availability Statement

Data will be made available on suitable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Ntombifuthi Q. Khumalo wrote the manuscript and all authors reviewed and edited the manuscript.

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Funding

No funding was received for this research.

Acknowledgements

The authors would like to thank their universities for the support provided.

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